SABD Project: Urban Mobility Analysis — Milestone M1

Research Note: Topic 1.2 — Daily Travel Times

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1. Scientific Problem & Research Question

This research investigates individual **Daily Travel Time** (*BTT*, *Budget Temps de Transport*) across the Grenoble metropolitan area using the 2010 Household Travel Survey (*EMD 2010*, Cerema standard). Grounded in Zahavi's conjecture (1980) and subsequent empirical evaluations (Crozet and Joly 2004, 2006; Mokhtarian and Chen 2004), we test whether travel expenditures exhibit behavioral invariance ("Marchetti constant" ≈ 1 h/day) or are structurally governed by residential sorting and spatial accessibility. We examine how housing characteristics (W) shape extreme commuting ($btt > 120$ min) versus immobility.

2. Unit of Analysis & Dataset Hierarchy

- **Natural Unit:** The **individual** (`allgreI`, $N = 8{,}861$ unique individuals nested within 3,802 households `id_men`). Among them, 7,425 are mobile ($nbd > 0$) while 1,436 are immobile ($nbd = 0$).
- **Pseudo-Replication & Grouped Splitting:** Household members share vehicle resources and residential constraints. To eliminate data leakage and anticonservative inference: (i) the 80/20 train/test split is strictly grouped by household (`id_men`) with `seed = 2026`; (ii) explanatory regressions employ cluster-robust standard errors (CRSE at `id_men` level) or mixed-effects models.

3. Target Variable Y: Daily Travel Time (`btt`)

- **Measurement & Distribution:** Y = btt is the aggregate daily duration (minutes) of all trips performed by an individual. In the mobile sample, Median = 65.0 min, Mean = 90.33 min, IQR = $[39, 110]$ min, and Max = 1,560 min. It displays strong right-skewness and declared survey rounding spikes ($5, 10, 15, 30$ min).
- **Immobility & Zero-Inflation:** Immobile individuals (`nbd == 0`) have `btt = NA` in raw records; we recode them to `btt_all = 0`. Because this zero-mass (16.2%) represents structural non-travel rather than missingness, we adopt a two-tier strategy:
  1. *Conditional Duration Model:* Estimation on active individuals ($btt > 0$) via $\log(btt)$ linear regression and Gamma GLM with log-link.
  2. *Two-Part (Hurdle) Model:* Logistic regression for mobility participation ($\Pr(nbd > 0)$) coupled with a truncated Gamma GLM for $\mathbb{E}[btt \mid btt > 0]$.

4. Imposed Variables (X, Z) & Focus Variable (W: Housing)

- X **(`jourdepl`):** Survey travel day (Monday–Friday). Controls for institutional intra-week rhythms (e.g. Wednesday school shifts in France, Friday early commuting). Formatted as a categorical factor at the individual level.
- Z **(`permis`):** Driver's license (0 = none, 1 = full license, 2 = accompanied driving). Strongly collinear with *age* (structural zeros for minors under 18). We recode category 2 into a dedicated factor level and test age-stratified interactions.
- **Focus W (Housing & Residential Environment):** A synthetic categorical indicator derived through unsupervised clustering (Gower distance + PAM / MCA followed by HCPC) fitted **exclusively on the training fold**.
  - *Features:* Housing typology (`TYPE_HAB`: single-family isolated, terraced, small/large collective building); tenure (`TYPE_OCU`: owner, private tenant, social tenant); residential parking friction (`PBM_STAT`: recurring parking tension at domicile); transit proximity (`Gare2`, `Gare5`).
  - *Zero Circularity:* W relies strictly on exogenous domicile and built-environment attributes, rigorously excluding trip distances (`D12`, `D13`), speeds, or route modes.

5. Methodological & Evaluation Roadmap

- **Predictive Protocol:** Grouped 5-fold cross-validation on the train set (80%). Baseline trivial model (sample mean/median) evaluated against Regularized Regression (Ridge/Lasso), Random Forest, and Gradient Boosting. Out-of-sample metrics: RMSE and MAE on both log-transformed and natural scales (minutes).
- **Explanatory Protocol:** Econometric Gamma GLM (log-link) estimating marginal effects, odds ratios, and cluster-robust inference.

References

Crozet, Yves, and Iragaël Joly. 2004. "Budget-Temps de Transport: Les Sociétés Urbaines Condamnées à La Vitesse?" *Les Cahiers Scientifiques Du Transport* 45: 11–35.

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Zahavi, Yacov, and Antti Talvitie. 1980. "Regularities in Travel Time and Money Expenditures." *Transportation Research Record* 750: 13–19.